

Tab 1

Figure 1:

Figure 2:

- IMDb-Wiki, so the loss is MSE (predicting age from faces. ~83 MSE means off by about 9 years)
- Nu = infeasible mass (%), how much target weight sits on groups the sampler under-serves
 - On IMDB, bigger beta means bigger nu (opposite of adult)
 - The sampling weights are proportional to i^{β} and the important identities are the low i ones, so bigger beta starves them more
 - Beta stops at 3 because that's the old mirrored cubic setup
- Solid line = measured (5 seeds $\pm$ 1 std), dashed = the floor (the best that method could ever reach (computed before training))
 - The dashed lines are the same in both panels, since the floor doesn't depend on the step size
 - Full (red) is flat at 83 because it sees every group every step
- Left (constant stepsize)
 - GAVOT loses at every beta
 - Even at beta = 0, where its floor is way lower, it loses, because of the noise from the big weights on rare groups
- Right (decaying stepsize, THE IMPORTANT ONE)
 - The solid lines sit on their dashed lines, so the prediction works
 - GAVOT wins big at beta = 0
 - The win shrinks as the dashed lines come together, flips around at beta = 1.5 to 2, and at beta = 3 GAVOT loses by 2.3
- Formatting issues
 - The legend doesn't say dashed = floor (only caption, might be ambiguous)
 - The Beta labels are crowded near beta = 1.5 and 2
 - Blue and orange overlap from beta = 1.5 on, so the crossover is hard to see

Figure 3:

- Difference between Cross entropy and MSE
 - Why is adult using one and IMDb using another?
 - Checked with Claude and it's lowk chill its cuz Adult has 2 classes of over or under 50k and MSE is designed for continuous numbers and predicted shi.
- Left
 - We are looking at the least represented group of the adult dataset, the other group
 - We are looking at what their cross entropy loss is for different paradigms and can confirm that full is the best obviously but GAVOT is solidly beating Group blind averaging
 - This metric is constructed using decaying learning rate
 - B and v are inversely proportional

- V is the amount of infeasible mass as a percent and is something that we set before the experiment, an independent variable
 - Why are we going until only that much v for the adult and not going more, why do the other figures have beta going until like 3
- We could theoretically put the exact final values for the graph but that might not be necessary
- Formatting issue: i think everything for v and beta should be organized like the bottom left metric, it looks the least crowded and most readable
- Right
 - The B could be placed in a better spot
 - We need a clarifying line somewhere that explicitly states what the different formulas are for B because clearly they are not being used the same
 - This metric shows the loss on the least represented group for imdb again with decaying step size
 - Its the 20 most important identities which are also the least observed once the B is greater than 0 what ever that means with this formula
 - GAVOT gets overtaken or rather it overtakes and does worse once the infeasible mass is like 70%. Up until that point it is hardily winning.

Tab 2

Adult dataset:

- The adult dataset is a census income dataset that, according to the survey on group fairness in federated learning, is the most widely used benchmark dataset for machine learning. The task of the dataset is to predict if an individual's income is above or below 50k
- We got it from the UCI machine learning repo
- I don't know exactly how we built 100 groups of 30; I gotta look at the paper
- The split being 85/10/3/1/1 is basically just the ratio of people in the various groups, with white having the most amount of data. This came down to just how the census was sampled at the time; it's from the 1990s.
- The target distribution is p , and it is set as .2, which is 1 divided by 5 for the 5 different groups
- We want to measure the cross-entropy loss of a baseline Federated Averaging Full, where all clients can participate at all times. In reality, due to client selection bias, all the clients may not be able to participate equally across the many rounds of training, and due to simple system constraints. We then compare two other methods that do not sample all the clients at once but rather take a set number
- What we want to measure is whether GAVOT will perform better, more equitably, than just doing group blind averaging in which a random subset of 3 clients will be taken. This just promotes the already baked-in bias, which is why GAVOT is necessary. We also experimented with a constant vs. a decaying learning rate derived from a simple formula that made the learning rate become, little by little, smaller over the course of the rounds. This made the loss go down even further and made the model we care about, GAVOT, even more accurate when it came to its task.
 - We actually are specifically measuring the tail 500 of the 3 different models so that we can have a single, averaged number that represents where the model approximately ended up after the training
- The significance of a good result would be that we were able to use the combination of GAVOT and a (decaying) gradient descent to have more equitable performance and not fall victim to the bias in using a regular thing like group blind averaging.

IMDB wiki

- Similarly, for the IMDb-Wiki dataset, we saw this as an established dataset for Machine learning
 - We got the dataset from Yuzhe Yang's Deep Imbalanced Regression GitHub repo (<https://github.com/YyzHarry/imbalanced-regression>) and used MSE because there were many different labels
 - The goal of this dataset was Human facial age estimation and gender classification.

Tab 3

Adult dataset

- It's a US Census dataset from 1994, with about 32,500 people. The job is to guess whether someone makes more or less than \$50K a year.
- Each person has things like age, education, job, hours worked and race. The numbers are scaled, and the categories are turned into yes/no columns.
- The model is simple logistic regression. It's scored with cross-entropy, a "how wrong were the guesses" score where lower is better. The learning rate is 0.1.
- How we built the 100 groups of 30:
 - Every group is 30 people, all of the same race.
 - The number of groups a race gets matches how common it is in the data: White is 85.5% of people, so 85 groups. Black 9.6% → 10, Asian 3.1% → 3, Amer-Indian 1% → 1, Other 0.8% → 1.
 - That's 100 groups and 3,000 people in total. They were picked with a fixed shuffle, so every run uses the same groups.
- What we want (the target): all 5 races should count equally, 1/5 each. So the single "Other" group is worth 0.2 all by itself, while each of the 85 White groups is worth only 0.2/85.
- The catch: under uniform sampling each step picks 3 of the 100 groups, and "Other" is only 1 of 100. So it shows up in only about 3% of steps, even though it's worth 20% of what we care about. That mismatch is the problem the paper fixes.

IMDb-Wiki dataset

- It's celebrity face photos, each labeled with the person's age. The job is to guess someone's age from their face.
- Age is a number, not a category, so the score is MSE (mean squared error: the average of how far off the guess was, squared). Lower is better. The dataset also has gender labels, but we don't use them.
- Where the data came from:
 - the photos are the original IMDb-Wiki dataset;
 - the age, photo and split list comes from Yuzhe Yang's "Deep Imbalanced Regression" GitHub repo;
 - we made the face features ourselves: each face goes through a pretrained image network (ResNet18) and gets squeezed down to 128 numbers.
- The model is simple linear regression, with learning rate 0.01.
- Groups: 100 people (identities), with 30 photos each (3,000 photos).
- Target: the people aren't equally important. Person 1 matters most, and it drops off fast after that, following $(101 - i)^3$.

How each run works (both datasets)

- Each step, we pick 3 of the 100 groups. The sampler doesn't pick fairly, and a skew knob β controls how unfair it is:
 - Adult: $\beta = 0$ means every group is equally likely to be picked, which really means races are picked by how common they are. $\beta = 1$ means groups are picked in line with how much they matter, which is the easy case.

- IMDb-Wiki: bigger β means the sampler picks the least important people more and more. The share of importance the sampler under-serves (v) grows from 31% to 88%.
- Each picked group trains on its own data for 5 small steps, and then the method decides how much each group's update counts.
- 4000 steps. The learning rate shrinks slowly over time, using $\eta/(1 + t/1000)$. We also ran a version where it stays constant.
- We ran everything 5 times with different random picks.
- What we report: the average loss over the last 500 steps, i.e. where the model ended up. Then we average that over the 5 runs.
- Welch's t : a test of whether a difference is real or just luck. Roughly, above 2–3 means it's real.

The methods we compare

- GAVOT (ours): uses the transport plan to decide how much each picked group counts.
- Group-blind averaging: every picked group counts 1/3. It ignores what we care about.
- Upsampling: boosts rare groups by (importance $\div$ how often they're seen), then rescales so the weights add to 1.
- Fixed multiplier: like upsampling, but without rescaling. The weights can add up to 15.6 instead of 1, which makes it unstable.
- LDS: reweights by how rare the age label is (IMDb-Wiki only).
- Full: sees all 100 groups every step. It's not a real option, since the sampler can't do this. It's just the best case to compare against.

The results (what we measure $\rightarrow$ why it matters $\rightarrow$ numbers)

1. Does GAVOT get the lowest loss overall? (Table 1a)

- Why it matters: it shows that fixing the weights actually gets you closer to what you wanted.
- Adult, uniform sampling: GAVOT 0.2269 vs group-blind 0.2479 ($t = 40$, very real). Upsampling 0.2288 is close but still worse ($t = 5.1$). Full 0.2073. So GAVOT closes 52% of the gap between group-blind and full.
- Adult, aligned sampling: everyone is basically tied, all within 0.0004 of full. That makes sense: when the sampler already matches what you want, there's nothing to fix.
- IMDb-Wiki, uniform: GAVOT 86.06 vs group-blind 91.63 ($t = 18$). Upsampling 86.74, LDS 94.28, full 82.95. That's 64% of the gap closed.
- IMDb-Wiki, aligned: GAVOT 84.28 vs group-blind 86.37 ($t = 11$). That's 61% of the gap closed.
- The fixed multiplier blows up: 0.670 on Adult, 8504 on IMDb-Wiki, and it diverges on the aligned cells.

2. When does GAVOT win and when does it lose? (Table 1b, IMDb-Wiki)

- Why it matters: it shows that you can tell in advance whether GAVOT will win, using a number called the alignment gain (Δ_{floor}).
- As the sampler gets more unfair (bigger β), the alignment gain shrinks: $6.99 \rightarrow 5.71 \rightarrow 3.67 \rightarrow 2.56 \rightarrow 1.62 \rightarrow 0.02$.

- GAVOT's actual win shrinks with it: $+5.57 \rightarrow +3.46 \rightarrow +2.05 \rightarrow +0.58 \rightarrow -0.03 \rightarrow -2.28$.
- The actual win is always about 1.4–2.3 smaller than the alignment gain, because GAVOT's bigger weights add noise.
- So GAVOT wins while the gain is big ($\beta \leq 1$), ties in the middle ($\beta = 1.5, 2$) and loses when the gain is gone ($\beta = 3$).
- Just looking at "whose weights are closer to the target" does not predict this. GAVOT's weights are closer at every β , even where it loses.

3. Does the learning rate matter?

- Why it matters: GAVOT puts big weights on rare groups, which makes training noisier. A shrinking learning rate smooths that noise out.
- With a constant learning rate, GAVOT loses at all 6 IMDb-Wiki settings, by up to 7.9 (116.4 vs 108.5).
- On Adult, GAVOT wins either way.
- These curves are in the GitHub repo, not the paper.

4. Do the rarest groups actually get better? (Fig. 2)

- Why it matters: this is the fairness part. These are the groups normal SGD ignores.
- Adult, race "Other": GAVOT 0.073 vs group-blind 0.119 (full 0.031). GAVOT is better at every skew.
- IMDb-Wiki, the 20 most important people: GAVOT is better while the sampler isn't too unfair (77.0 vs 86.4 at the mildest setting; full 72.2). It ties in the middle and loses at the three most unfair settings.

5. Does adding CVaR help?

- Why it matters: CVaR is a safety guarantee on the target, not a way to lower the average loss.
- We tried 100 combinations of its two settings (α, γ). The lowest average loss was always at, or basically at, $\gamma = 0$, i.e. CVaR turned off.
- It does help the worst-off race on Adult a little: 0.3624 (at $\alpha = 0.2, \gamma = 0.3$) vs GAVOT 0.3679 vs group-blind 0.3994. That setting isn't one the theory guarantees, though.
- It never helps the single rarest group, because CVaR can only reweight groups that are already in the batch.

Writing

The adult dataset \cite{uci_adult} is a 1994 US census income dataset on about 32,500 people. The task is to predict whether an individual's income is above or below \$50,000 a year using features like age, education, job, and marital status. We use a logistic regression model trained with cross-entropy loss at a learning rate of 0.1. The adult dataset is divided into 5 races: White, Black, Asian-Pacific-Islander, American-Indian-Eskimo, and Other. The ratio of the distribution is approximately 85:10:3:1:1, respectively. We use a subset of 100 groups of 30 people, equivalent to a total of 3,000 people. This matches our IMDb-Wiki setup of 100 people with 30 photos each, and small groups keep our many training runs fast. Each group is all one race, and we give each race a number of groups that follows that ratio: 85 White groups, 10 Black groups, etc. Each model step picks 3 groups, and since Other has only 1 group out of 100, it lands in a batch only about 3% of the time when groups are picked uniformly. The target weights the five races equally, $\frac{1}{5}$ each. Since Other has only one group, that group carries the full 0.2 (20%). Our paper aims to solve the discrepancy between Other being seen 3% of the time when it should count 20%.

The IMDb-Wiki dataset \cite{rothe2018imdbwiki} is a collection of over 500,000 celebrity faces from IMDb and Wikipedia with age and gender labels. We only use age labels. The task is to predict a person's age from a photo of their face. We turn each face into 128 numbers using a pretrained ResNet-18 network and then PCA. We use a linear regression model trained with MSE loss at a learning rate of 0.01. MSE is used since age is a number, so this is a regression task. The MSE itself is in years²: e.g., an MSE of about 86 years² means a typical error of about 9 years. The IMDb-Wiki subset we used contains 100 people, each with 30 photos, for a total of 3,000 photos. Each group is one person, so the same 100 x 30 shape arises as it did in Adult. We choose a steeply skewed target so that a few people dominate. Person i gets importance proportional to $(101 - i)^3$, so person 1 matters the most and importance drops off quickly. To put it in proportion, the top 20 people carry about 60% of the target.

The sampler picks 3 groups each step, and the picking is uneven by design. A skew knob β sets how uneven the picking is, which simulates data sources where some groups show up far less often than they matter. ν is infeasible mass, which is the share of the target that belongs to groups seen less often than they matter. For the Adult dataset, we test β values of 0, 0.25, 0.5, 0.75, and 1, running in even steps from uniform picking (0) to fully aligned picking (1). $\beta = 0$ means every group is equally likely to be picked, so races show up by how common they are, and Other is in about 3% of batches. $\beta = 1$ means groups are picked in line with the target. That is, each group's chance of being picked is proportional to how much it matters, so every race shows up about equally often. On Adult, ν goes from 60% to 0. For the IMDb-Wiki dataset, a bigger β makes the sampler favor the least important people more, which makes the problem harder. We test $\beta = 0, 0.5, 1, 1.5, 2, 3$, running up to 3, the most extreme mismatch, so the sweep shows where transport stops winning. On IMDb-Wiki, ν goes from 31% to 88%. We also test an aligned version, where people are picked roughly in line with the target, so $\nu = 0$. Each picked group trains on its own 30 examples for 5 steps, then the method

combines the results using its weights. The learning rate shrinks over time following the decay formula $lr/(1 + t/1000)$. Here t is the step number, so the rate is halved by step 1000 and is a fifth of its starting value by step 4000. This simple standard schedule keeps early learning fast and smooths out the extra noise that GAVOT's big weights on rare groups create. Each run lasts 4000 steps, and we do 5 runs with different random seeds. We report the average loss over the last 500 steps, averaged over the 5 runs. Welch's t-statistic checks whether a difference is real (roughly, $|t| > 3$ means clearly real).

- Group blind is basically that when you are randomly selecting a subset of clients, you don't care about what race they come from. This is not specific to race; that is just for the adult dataset. It means that you don't care about what group the client belongs to when you select them, e.g., whether they are overrepresented or underrepresented.
- Full access means that you have full access to every single client and their data during every single round of training. Due to real-world constraints, this is often not possible, but as a theoretical scenario, it functions as the baseline for all the other models.
- GAVOT is taking a different approach by solving a "transport problem". It basically constructs a table that reweights all the groups so that the rows add up to the target p and the columns match their probability of being drawn, which I think is π_i ? [Herlock I need you to check this bit since its been a bit since i took all the notes and spent a couple hours understanding them] Using the Sinkhorn algorithm it keeps bouncing between rescaling the rows and then the columns and back again so that they get closer and closer to matching but i think for an infeasible scenario they will never match
- Upsampling put simply means to reweight drawn groups by duplicating rarely seen groups and giving them more weight.
- LDS/Label Distribution Smoothing is a bias mitigation technique in which you ignore the p and you ignore availability; instead, you focus on
- Fixed multiplier means that all of the drawn groups before the training are assigned a weight, but there is no renormalization. The weights dont sum to 1 and act like a much larger step size, leading to divergence.

We measured the overall target loss for each of the methods in order to demonstrate that adjusting the weights gets the training closer to the baseline of full access and makes the training more equitable. In Adult uniform, the GAVOT tail 500 was .2073 vs group-blind .2479 at $t=40$, t being the welch t series; the full baseline is .2073 and GAVOT closes 52% of that gap. In IMDb Wiki,

With a constant stepsize GAVOT loses on IMDb-Wiki every time when measuring MSE, with the worst case being 116.4 vs 108.5 at $\beta = 3$. On Adult however, it wins in both constant and decaying schedules, the curves are in the Github repo.

The first question is whether correcting the weighting lowers the target risk, and Table 1a shows that it does. On Adult under uniform sampling, GAVOT reaches 0.2269 against 0.2479 for group-blind averaging ($t = 40$), closing 52% of the gap to full access. On IMDB-Wiki it closes 64% of the gap under uniform sampling (86.06 against 91.63, $t = 18$) and 61% under the aligned sampler (84.28 against 86.37, $t = 11$). Upsampling, which renormalizes batch by batch, recovers most of this gain but not all ($t = 5.1$ on Adult, 2.3 on IMDB-Wiki), and LDS, which ignores availability, trails group-blind averaging. On aligned Adult there is nothing to correct, and every normalized rule lies within 0.0004 of full access. The unnormalized fixed multiplier, whose batch weights sum to as much as 15.6, reaches 0.670 and 8504 on the uniform cells and diverges on the aligned ones.

As β grows, the alignment gain (the most that correcting the weighting could save with perfect training) falls from 6.99 to 0.02, and the measured gain stays below it by 1.4 to 2.3, the price of the larger and noisier weights. Transport therefore wins while the alignment gain clearly exceeds this price ($\beta \leq 1$), ties at $\beta = 1.5$ and 2, and loses at $\beta = 3$ ($t = -7.1$), even though its weighting is closer to the target at every skew, so distance to the target does not predict the winner. With a constant step size, the price grows, and transport loses at all six skews, by up to 7.9 in MSE, whereas on Adult it wins at every skew under both step sizes.

On Adult, transport lowers the cross-entropy of race Other at every skew, to 0.073 against 0.119 for group-blind averaging under uniform sampling (full access baseline reaches 0.031). On IMDB-Wiki, the 20 most important identities improve while $\nu \leq 0.59$ (77.0 against 86.4 at $\nu = 0.31$), tie at $\nu = 0.70$, and lose at the three most skewed samplers, where transport puts the largest weights on them and the added noise dominates.

The CVaR layer provides a guarantee rather than a lower loss. Over a 10×10 grid of (α , γ), the target loss is lowest at, or within noise of, $\gamma = 0$, where the layer is off, in every instance, which is the price that Theorem 4(ii) predicts. The layer lowers the loss of the worst-off race on uniform Adult to 0.3624 at $(\alpha, \gamma) = (0.2, 0.3)$, a point outside the certified region, against 0.3679 for GAVOT and 0.3994 for group-blind averaging, but it never helps the least represented group, since it can reweight only the groups present in a batch. The per-skew curves, the constant-stepsizes runs, the full grid and the code are in the repository.